

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Tech. Examination (EE)

May 2018

EE 772: Restructuring & Deregulation of Power Systems

Date: 10.05.2018, Thursday

Time: 1.30 to 4.30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q – 1** Define congestion and explain its disadvantages. [07]
- Q - 2 (a)** Explain drawbacks of conventional power system and write advantages of restructured power system. [07]
- (b)** Explain Total Transfer Capability (TTC) and its associated terms. [07]

OR

- Q - 2 (a)** Explain DC Power Transfer Distribution Factor for determining system ATC. [07]
- (b)** Explain Single auction and double auction models. [07]
- Q – 3** Answer any two questions from the following: (each question carry seven marks) [14]
- (a)** Explain active power flow tracing algorithm using proportional sharing approach.
- (b)** Different types of ancillary services.
- (c)** Explain the concept of MCP and LMP

SECTION – II

- Q - 4** Explain the role of Independent system operator in the restructured environment. [07]
- Q - 5 (a)** Different types of pricing policies of a restructured power system. [07]
- (b)** Explain OPF formulation for ATC maximization subject to various constraints [07]

OR

- Q - 5 (a)** Explain Day-ahead market. [07]
- (b)** Explain the rules of Intra-zonal OR Inter-zonal congestion management. [07]

Q - 6 Answer any two questions from the following: (each question carry seven marks) [14]

- (a)** Draw and Explain different types of restructured markets.
- (b)** Explain voltage distribution factor and line outage voltage distribution factor
- (c)** Explain Loss of opportunity cost of a synchronous generator.
